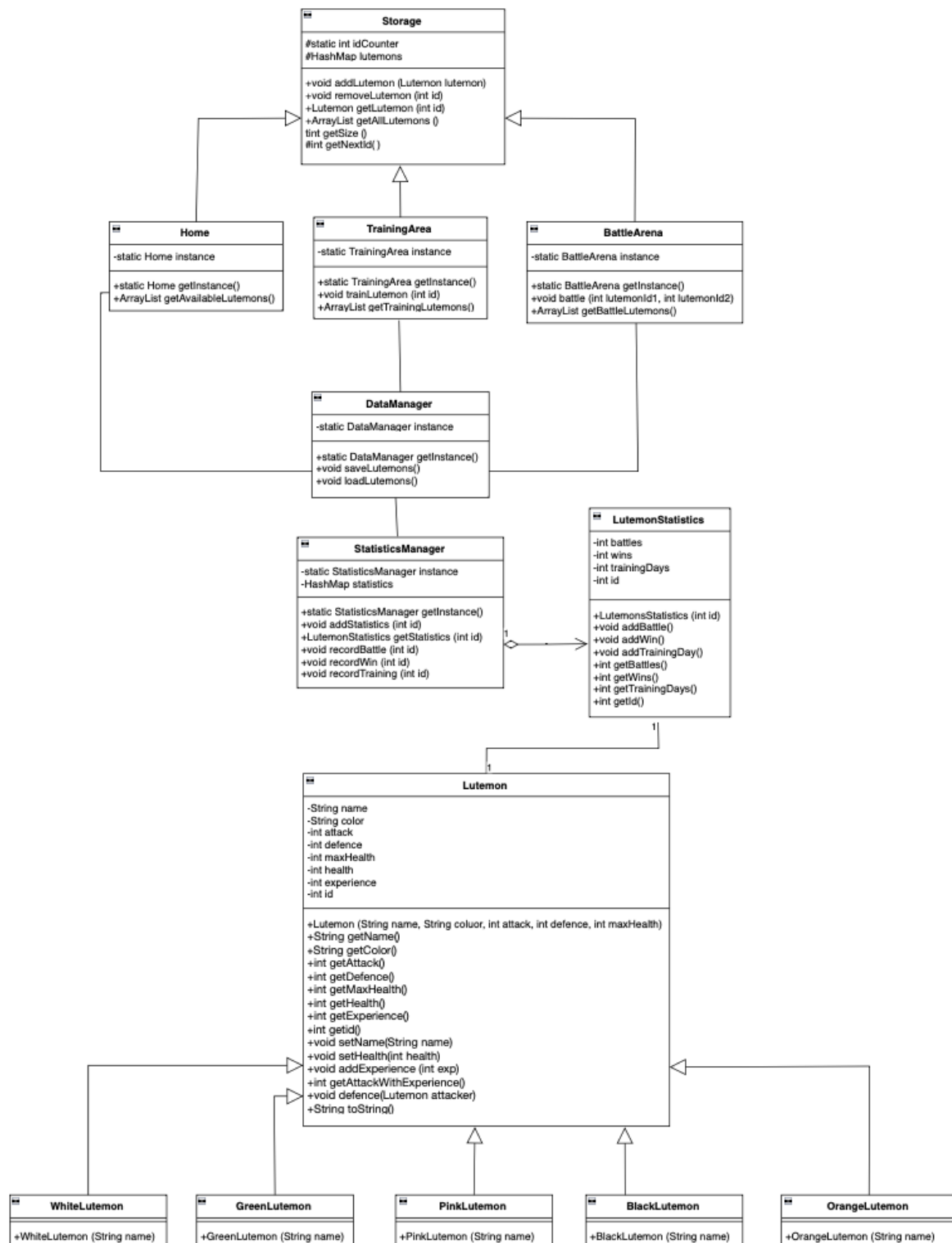


Project Work Documentation

Project Name: Lutemon World

Project Author: Vera Stojcheva, 001164725 (individual work, group 007)

UML Class Diagram



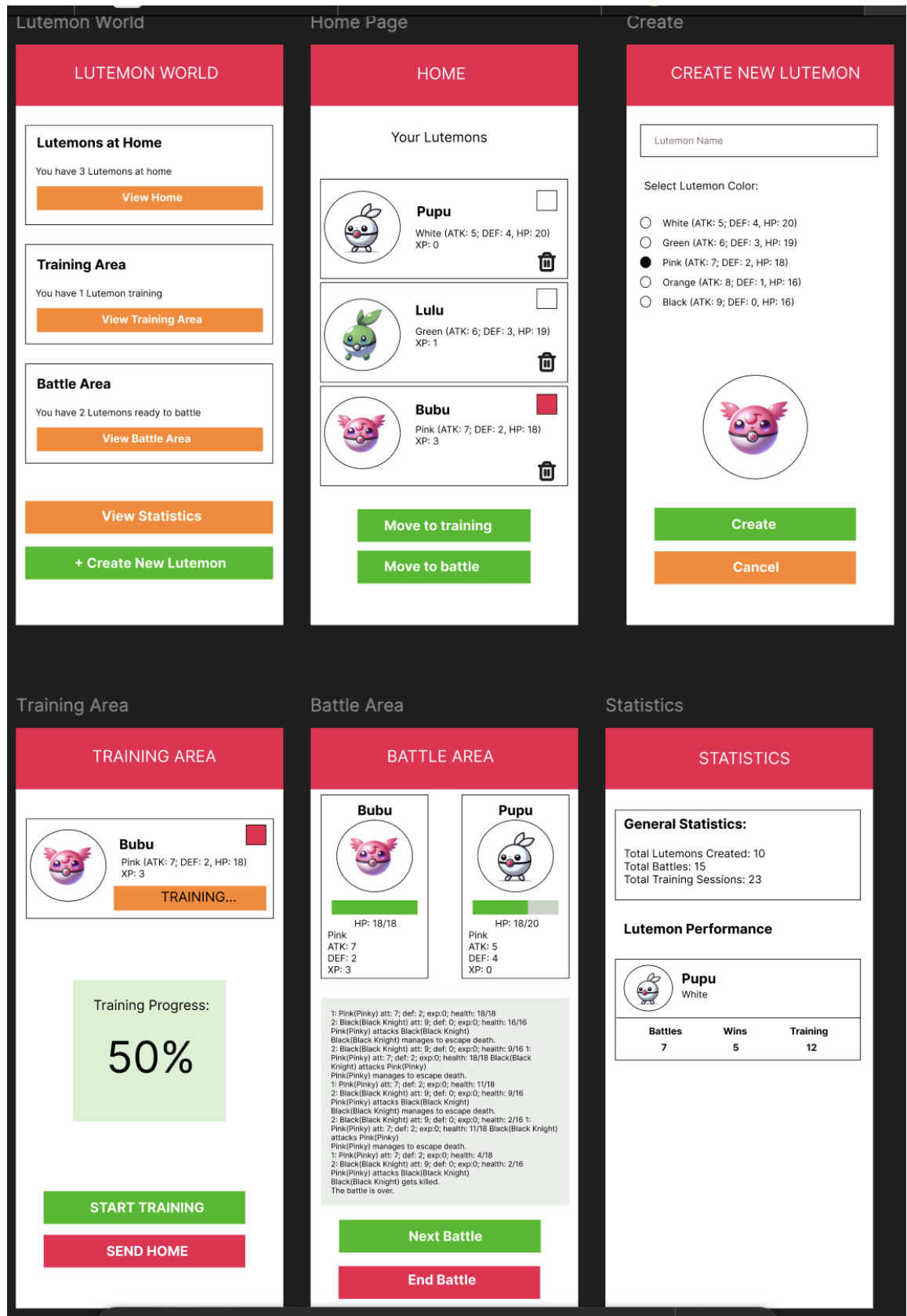
Implemented Features

Requirement	Description	Points
Object-Oriented Code	The program is coded according to the object-oriented paradigm.	Required
Code in English	All code and comments must be in English. However, documentaion and UI elements can be in Finnish.	Required
Android Compability	The program must work on various Android phones and be coded in Java using Android Studio.	Required
Basic Functonality	The program must implement all the features defined above so that users can interact with Lutemons.	Required
Documentation	The final submission must include documentation (PDF or GitHub .md file) containing: class diagram (excluding UI classes like Activities), division of work among team members, implemented features, and a general project description.	Required 13
Lutemons Have Images	Different Lutemons are visualized with unique images.	+1
RecyclerView	Uses the RecyclerView component to list Lutemons and their details.	+1
Statistics	Track how each Lutemon has performed (number of battles, victories, training days, etc.).	+2
No Death	Instead of dying, a defeated Lutemon returns to its initial state. Lost battles can also be tracked in statistics.	+2
Delete Lutemon	Option to also delete a Lutemon from the “Create Lutemon” page, with speified icon.	+2
Battle Visualization	By default, a textual description of the battle is displayed in a TextView. Improved visalization by dynamically adding visual battle effects, such as an attacking Lutemon’s image displaying a sword pointing at the defender.	+2
Fragments	Use fragments meaningfully in the application.	+2
Randomness in Battles	Some randomness to battles, e.g., <code>attack + Math.random() * 3</code> .	+1
Statistics Visualization	Visualize game statistics in some way (e.g., using AnyChart-Android).	+3

Total: 29

UI Design in Figma

Link: <https://www.figma.com/design/TkT9GFxhQ4PDpMUyEfPM0j/LutemonWorldApp?node-id=0-1&t=oeppmrklwdSOAlaz-1>



Final UI design and its demonstration is available in the video recording.

Funcitonality Description

The app consists of **six navigable screens**, each accessible via buttons from the main menu.

1. Lutemon World (Main Menu):

Serves as the central navigation hub. From here, users can access all the main features of the app: Create Lutemon, View Home, Training Area, Battle Area, and Statistics.

2. Create Lutemon Page:

The app begins with the user creating a new Lutemon by giving it a name and selecting a color (which determines its stats). Upon creation, the Lutemon appears on the Home screen as a card and is included in the statistics tracking system.

3. Home Page:

This is where all created Lutemons are listed- they are **never removed** from this view unless manually deleted.

Users can select one or multiple Lutemons and:

- Move them to **Training**
- Move them to **Battle**
- **Delete** them permanently

The number of Lutemons and their current state (training or battling) is dynamically reflected here and on the Main Menu.

4. Training Area Page:

Lutemons flagged for training are shown here. Users can start training, which gradually loads and awards **experience points (XP)**. There's no limitaiton on how much a Lutemon can be trained and the XP gained are also shown on the Home screen.

5. Battle Arena Page:

Two Lutemons can be selected for battle. Battles are turn-based and feature:

- A **textual play-by-play** of the fight
- A **sword graphic** showing who is attacking

The winner gains XP. The losing Lutemon is not deleted- it simply resets its stats (XP and health). It can be trained again and reused.

6. Statistics Page:

All Lutemons' progress is summarized here:

- Number of battles fought
- Victories
- Training sessions

These are updated live as actions happen throughout the app.

Reflection

Creating this app was an interesting and educational experience. I began by designing the initial user interface in Figma, which helped me visualize the concept and better understand the app's functionality before writing any code.

From there, I encountered many ups and downs while developing the app- especially when creating the classes and organizing the logic and methods. One of the most challenging parts was ensuring that when a Lutemon is moved to either training or battle, it is not removed from the Home screen, and that it also correctly appears in the respective screen. Another complex task was building the logic to update all the statistics in real-time, across the main Lutemon World screen, the Home screen, and the Statistics screen- without inconsistencies. Throughout the project, I used various tools and resources to support my learning. Online forums often had insightful discussions that helped me debug problems. I also used a few YouTube tutorials, and AI tools, which explained issues clearly and offered options for resolving them.

Even though this app may not serve a highly practical purpose in a real-world setting, it fulfilled my personal goal- to experience what it's like to create and develop an entire app from scratch on my own. It has been a valuable learning journey in both Android development and problem-solving.